

## OVERVIEW

8 study guide sites covering Modbus, OPC UA, DNP3, IEC 61131-3, PID Controllers, SEL RTAC, Ignition SCADA, and Wireshark. Each guide is a React/Vite SPA deployed to GitHub Pages, with quizzes, flashcards, a Code Lab, and a downloadable course PDF. The hub landing page ([scada-hub.vercel.app](https://scada-hub.vercel.app)) links all 8 guides.

---

## FEATURE WORK

- Quiz Question Hints** HIGH  
Add a hint field to every quiz question in all 8 guides' quizzes.js data files, showing WHERE the answer can be found (chapter name, section heading, flashcard ID). Add a "Hint" button to Quiz.jsx that reveals the source on click. Scope: 8 guides × 600–900 questions each.
  - Certification Relevance Sidebar Panel** MED  
Build a new sidebar button per guide (modal or slide-out panel) containing rich certification content: cert name, exam body, link, description, and which course chapters map to which exam domains. Content removed from the syllabus PDF; this is its permanent home in the web app. Scope: 8 guides.
  - DNP3 Study Guide LaTeX PDF** MED  
Write, compile, and deploy the DNP3 full-length study guide PDF (equivalent to the Modbus PDF). Place at site/public/study\_guide.pdf. Current sidebar PDF button returns 404. Scope: DNP3 guide only.
  - Wrong-Answer Tracking + Review Missed Mode** MED  
Store which question IDs a user missed per chapter (Supabase already wired). Add a "Review Missed" quiz mode that surfaces only previously incorrect questions. Scope: all 8 guides + Supabase schema.
  - Spaced Repetition on Flashcards** LOW  
Replace the current random shuffle with the SM-2 algorithm (~150 lines JS). Cards missed come back sooner; mastered cards surface less often. Scope: all 8 guides Flashcards.jsx.
  - Timed Exam Mode** LOW  
Hub-level: 60 questions, 90 minutes, randomized across all guides in a topic area (OT protocols, ICS security, etc.). Supabase for scoring and leaderboard. Scope: scada-hub + Supabase.
  - Cross-Guide "See Also" Callouts** LOW  
Add "See Also" callouts within chapters that link to related chapters in other guides (e.g., Wireshark Modbus chapter → Modbus guide, OPC UA Security → Wireshark OPC UA chapter). Scope: all 8 guides content pages.
- 

## CONTENT GAPS — SYLLABUS VS. COURSEWORK

- Modbus: Prerequisites + ModRSSim2 + TCP Flashcards** MED  
(a) Add prerequisite statement to Home.jsx / Intro chapter. (b) Mention ModRSSim2 in the TCP or Required Software section of the web guide. (c) Expand TCP chapter flashcards (currently 3; syllabus implies 5+, covering transaction ID pipelining and Unit ID in gateways).
- OPC UA: Sessions/Channels Chapter + Discovery Section** MED  
(a) Add a dedicated Sessions & Channels section to Architecture.jsx: token renewal, reconnection retry, ActivateSession flow. (b) Add a Discovery section covering GetEndpoints, FindServers, GDS pattern, certificate trust flow. (c) Introduce open62541 in at least one chapter page (currently only referenced in quiz questions).
- IEC 61131-3: PLCopen FB Library + Vendor Portability** MED  
(a) Add a section on PLCopen standard FB signatures (TON/TOF/CTU), instance management, and library navigation. (b) Add a vendor portability section on migration barriers across Siemens/Beckhoff/SEL implementations. (c) Add a table explaining which IEC version applies to which platforms.
- PID: Scilab Lab Exercises + Laplace Prerequisites** MED  
(a) Add Scilab step-test simulation exercises to the Process Characterization chapter. (b) Add a note to the Digital Implementation chapter clarifying the math prerequisites (Laplace/transfer functions) so students without calculus know what to skip or review first.
- DNP3: Prerequisites Statement in Web Guide** LOW  
Add a brief prerequisite note to Home.jsx or the Intro chapter stating that basic TCP/IP and Modbus familiarity is assumed. All 10 topics are fully covered; this is the only gap found.
- RTAC: Modbus Gateway Chapter** MED  
Add a dedicated Modbus Gateway chapter (or major section) covering: DNP3-to-Modbus point mapping, register layout design, data type conversion rules, and translation failure modes. Syllabus Topic 4 ("Modbus Gateway") has no corresponding web chapter.

Also expand I/O module selection and integrate SEL-5033 virtual RTAC into the lab exercise.

- 14. **Ignition: OPC-UA Integration Chapter Depth** MED  
Audit `opcUa.jsx` against syllabus Topic 3 (OPC-UA Integration). Verify PLC-to-live-tags workflow and Ignition-as-OPC-UA-server config are fully covered. Add SQL prerequisite note to `Home.jsx`. (*Audit pending.*)
  - 15. **Wireshark: SSH Remote Capture + tcpreplay** LOW  
Syllabus Topic 9 (Advanced Techniques) lists SSH remote capture and tcpreplay. Verify both are in `Advanced.jsx`. Add content where missing. (*Audit pending.*)
- 

## BUGS & INFRASTRUCTURE

- 16. **Vercel Re-Deploy All 8 Guides** HIGH  
Free-tier 100-deploy/day limit was hit on 2026-05-15. All code is on GitHub; run `vercel --prod --yes` from each guide root to sync the latest builds to Vercel. Affected: all 8 guides.
  - 17. **Auth: Duplicate Email Signup Error** MED  
Registering with an existing email shows a generic Supabase error. Show a clear "account exists" message with a "Reset password" link. Add reset-password flow triggered from the register form error state. Scope: `scada-hub AuthPage.jsx`.
  - 18. **Quiz Reports RLS — Supabase** MED  
Any logged-in user can currently read any other user's quiz history. Add Supabase row-level security (RLS) policies on the `quiz_results` table: users see only their own records; admin account has a bypass policy.
  - 19. **Mobile UX Audit** MED  
Overall mobile layout is unfriendly across all 8 guides and the hub. Full audit needed: text sizes, tap targets, sidebar behavior, `LearningPath` horizontal scroll on small screens.
  - 20. **Supabase Email Confirmation Redirect** LOW  
Clicking the confirmation email link sends users to a broken URL. Fix: update Site URL and Redirect URLs in Supabase Auth settings to point to `scada-hub.vercel.app`.
  - 21. **LearningPath Hover Scale Bug** LOW  
Hovering a step card triggers `whileHover={scale: 1.03}` — card enlarges and text becomes hard to read. Reduce or remove the scale transform. Scope: `scada-hub/site/src/components/LearningPath.jsx`.
  - 22. **RTAC PDF Layout Overhaul** LOW  
RTAC study guide PDF has bad spacing, orphaned headings, and an intro style inconsistent with the Modbus PDF template. Needs a formatting pass to align with the Modbus LaTeX template.
  - 23. **OG Image Quality Improvement** LOW  
Current OG images use Python PIL with the system Helvetica font — functional but basic. Replace with a custom SVG design or Vercel's `@vercel/og` for server-side image generation at deploy time. Scope: `scada-hub public/og-image.png` + all 8 guide repos.
- 

## UI / BRANDING

- 24. **Login Page: Company A Brand Colors** MED  
The SCADA Hub login/auth page (`ScadaBackground.jsx` + `AuthPage.jsx`) uses orange as its primary accent. Adjust to a specific hex color value provided by Company A. Scope: `scada-hub/site/src/components/ScadaBackground.jsx`, `AuthPage.jsx`. Collect the target hex value before implementation.
  - 25. **Login Page: Floating Telemetry Values** MED  
On the animated SCADA login background, the live telemetry tickers (PV kW, SOC%, RTT ms, MVA values) overlap with the node icons on some screen sizes. Fix: (a) move ticker boxes so they do not cover the icon, and (b) animate them with a slow float (gentle vertical oscillation) to give the background more life. Scope: `ScadaBackground.jsx` node layout and ticker positioning logic.
  - 26. **All Courses Sidebar Panel — Design Polish** LOW  
The "All Courses" collapsible nav item was added to all 8 guide sidebars with guide icons and Vercel URLs. Future polish: (a) highlight the current guide in the dropdown so users know which guide they are on, (b) consider adding the guide's chapter count or question count as a subtitle under each guide name.
- 

## PENDING AUDITS

- 27. **Ignition SCADA: Syllabus vs. Coursework Audit** MED  
Audit `ignition-study-guide` chapter pages against the Ignition syllabus (Course 7 of 8). Verify OPC-UA Integration chapter depth (syllabus Topic 3), SQL prerequisite note, and that all 10 syllabus topics have corresponding web content. Document gaps in the devlog and add fix items to punchlist.
- 28. **Wireshark: Syllabus vs. Coursework Audit** MED  
Audit `wireshark-study-guide` chapter pages against the Wireshark syllabus (Course 8 of 8). Specifically verify: SSH remote capture and tcpreplay coverage in `Advanced.jsx` (syllabus Topic 9), and that all 10 topics have web content. Document gaps in the devlog.

## 29. Vercel Re-Deploy — All 9 Projects

HIGH

Vercel free-tier 100-deploy/day limit was hit on 2026-05-16. All code is current on GitHub. Run the following from each project root (build with empty GITHUB\_PAGES so base is /): `cd site && GITHUB_PAGES= npm run build && cd .. && vercel --prod --yes`  
Projects: modbus, opcua, dnp3, iec61131, pid, rtac, ignition, wireshark, scada-hub.

## 30. Troubleshooting Lesson Plan LaTeX PDF

MED

Build a multi-chapter LaTeX PDF documenting real debugging sessions as field-engineer lesson plans. Each chapter covers one class of bug or architectural decision: symptom, what was checked, root cause, fix, and what to watch for next time. Example chapter: "Missing Icon Import Crashes React App" (Ignition Zap ReferenceError). Generate at the end of each session and accumulate chapters over time. Source lives in `scada-hub/lesson-plans/`. Distinct from the per-guide study guide PDFs — this is a *developer training document*.